

Vessel Analog Signal 1 Malfunction

Symptoms

- The vessel sensor on the OEM application is **not** communicating with DCU410 unit.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The vessel sensor inputs on the customer interface box (CIB) are used by the OEM. The input to the DCU410 measures these values from the sensors.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box wiring.	
	STEP 1A. Check the DCU410 unit display for faults.	
	STEP 1B. Check the vessel temperature sensor 1 signal, return, and return 2 wires for an open.	
	STEP 1CB. Check the vessel temperature sensor 1 signal, return, and return 2 wires for a wire-to-wire short.	
	STEP 1D. Check the vessel temperature sensor 1 signal wire for a short to ground.	
STEP 2.	Check the OEM wiring harness.	

STEPS		SPECIFICATIONS
	STEP 2A. Check the vessel temperature sensor 1 signal, return, and return 2 wires for an open.	
	STEP 2B. Check the vessel temperature sensor 1 signal, return, and return 2 wires for a wire-to-wire short.	
	STEP 2C. Check the vessel temperature sensor 1 signal wire for a short to ground.	

STEP 1. Check the customer interface box wiring.

STEP 1A. Check the DCU410 unit display for faults.

Conditions :

- Locate the DCU410 unit display.

Action	Specification/Repair	Next Step
Check the DCU410 unit display for faults.	DCU410 unit indicates fault(s)? YES	1B
	DCU410 unit indicates fault(s)? NO	2A

STEP 1B. Check the vessel temperature sensor 1 signal, return, and return 2 wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the vessel temperature sensor 1 signal and return wires from the DCU410 unit and X4 connection.

Action	Specification/Repair	Next Step
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Check the vessel temperature sensor 1 signal, return, and return 2 wires for an open. • Place one test lead on the vessel temperature sensor 1 signal wire at the DCU410 unit. • Place the other test lead on the vessel temperature sensor 1 signal wire at the X4 connection. • Place one test lead on the vessel temperature sensor 1 return wire at the DCU410 unit. • Place the other test lead on the vessel temperature sensor 1 return wire at the X4 connection. • Place one test lead on the vessel temperature sensor 1 return 2 wire at the DCU410 unit. • Place the other test lead on the vessel temperature sensor return 2 wire at the X4 connection. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1C
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
STEP 1C. Check the vessel temperature sensor 1 signal, return, and return 2 wires for a wire-to-wire short.		
Conditions : • Open the customer interface box.		
Action	Specification/Repair	Next Step

Check the vessel temperature sensor 1 signal, return, and return 2 wires for a wire-to-wire short. • Place one test lead on the vessel temperature sensor 1 signal wire at the DCU410 unit. • Place the other test lead on all other wires at the DCU410 unit. • Place one test lead on the vessel temperature sensor 1 return wire at the DCU410 unit. • Place the other test lead on all other wires at the DCU410 unit. • Place one test lead on the vessel temperature sensor 1 return 2 wire at the DCU410 unit. • Place the other test lead on all other wires at the DCU410 unit.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? NO	1D

STEP 1D. Check the vessel temperature sensor 1 signal wire for a short to ground.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step
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Check the vessel temperature sensor 1 signal wire for a short to ground. - Place one test lead on the vessel temperature sensor 1 signal wire at the DCU410 unit. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	Repair complete
	Less than 10 ohms? NO	2A

STEP 2. Check the OEM wiring harness.

STEP 2A. Check the vessel temperature sensor 1 signal, return, and return 2 wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the vessel temperature sensor 1 signal, return, and return 2 wires at the X4 connection.
- Disconnect the vessel temperature sensor 1 sensor connector.

Action	Specification/Repair	Next Step

Check the vessel temperature sensor 1 signal, return, and return 2 wires for an open. • Place one test lead on the vessel temperature sensor 1 signal wire at the X4 connection. • Place the other test lead on the vessel temperature sensor 1 signal pin at the sensor connector. • Place one test lead on the vessel temperature sensor 1 return wire at the X4 connection. • Place the other test lead on the vessel temperature sensor 1 return pin at the sensor connector. • Place one test lead on the vessel temperature sensor 1 return 2 wire at the X4 connection. • Place the other test lead on the vessel temperature sensor 1 return 2 pin at the sensor connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	2B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to the OEM service manual.	Repair complete

STEP 2B. Check the vessel temperature sensor 1 signal, return, and return 2 wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the vessel temperature sensor 1 signal, return, and return 2 wires at the X4 connection.
- Disconnect the vessel temperature sensor 1 sensor connector.

Action	Specification/Repair	Next Step
Check the vessel temperature sensor 1 signal, return, and return 2 wires for a wire-to-wire short. • Place one test lead on the vessel temperature sensor 1 signal wire at the X4 connection. • Place the other test lead on all other wires at the X4 connection. • Place one test lead on the vessel temperature sensor 1 return wire at the X4 connection. • Place the other test lead on all other wires at the X4 connection. • Place one test lead on the vessel temperature sensor 1 return 2 wire at the X4 connection. • Place the other test lead on all other wires at the X4 connection. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to the OEM service manual.	Repair complete
	Less than 10 ohms? NO	2C

STEP 2C. Check the vessel temperature sensor 1 signal wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the vessel temperature sensor 1 signal wire at the X4 connection.
- Disconnect the vessel temperature sensor 1 sensor connector.

Action	Specification/Repair	Next Step
Check the vessel temperature sensor 1 signal wire for a short to ground. • Place one test lead on the vessel temperature sensor 1 signal wire at the X4 connection. • Place the other test lead on engine ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to the OEM service manual.	Repair complete
	Less than 10 ohms? NO Repair : Replace the vessel temperature sensor 1 sensor. Refer to the OEM service manual or contact a Cummins® Authorized Repair Location.	Repair complete